

TOSUtily Database – Utility Billing & Usage Data Model

This document describes the logical relationships between tables contained in the TOSUtily SQL dump. The schema does not enforce foreign keys; relationships are inferred from naming conventions and shared key columns.

1. Customers & Utility Accounts

ub_customers (Customer master)

- a_customer (Customer ID)

utactmst (Utility account master)

- utacm_key (Account key)
- utacm_account (Account number)

utactcid (Customer–Account link)

- utacd_cid → ub_customers.a_customer
- utacd_utacm_key → utactmst.utacm_key
- utacd_account → utactmst.utacm_account

2. Services, Meters & Usage

utsvcmst (Service master per account)

- utsvm_key
- utsvm_utacd_key → utactcid.utacd_key

utmtrmst_table (Meter inventory)

- utmtm_key

utmtrdtl (Meter details)

- utmtd_key
- utmtd_utmtm_key → utmtrmst_table.utmtm_key

utsvcmtr (Service–meter linkage / readings)

- utsmt_utsvm_key → utsvcmst.utsvm_key
- utsmt_utmtd_key → utmtrdtl.utmtd_key

3. Utility Billing

utbilhdr (Utility bill header)

- utblh_key
- utblh_utacd_key → utactcid.utacd_key

utbilsvc (Billed services)

- utbls_key
- utbls_utblh_key → utbilhdr.utblh_key
- utbls_utsvm_key → utsvcmst.utsvm_key

utbilmtr (Billed meter usage)

- utbmt_utbls_key → utbilsvc.utbls_key
- utbmt_utmtd_key → utmtrdtl.utmtd_key

4. Adjustments / Rebills

utadjhdr (Adjustment header)

- utalh_key
- utalh_utblh_key → utbilhdr.utblh_key
- utalh_utacd_key → utactcid.utacd_key

utadjsvc (Adjustment services)

- utals_key
- utals_utalh_key → utadjhdr.utalh_key
- utals_utsvm_key → utsvcmst.utsvm_key

utadjmtr (Adjustment usage)

- utamt_utals_key → utadjsvc.utals_key

5. Accounts Receivable (AR) Integration

ut_ar_reference (Utility → AR bridge)

- a_account, a_service_code, a_service_sequence
- a_bill_run, a_commitment
- a_ar_customer_cid
- a_bill_year, a_bill_number, a_ar_category

ar_customer (AR customer master)

- a_ar_customer_cid

ar_all_bills / ub_bill_header / ub_bill_detail

- Linked via (a_bill_year, a_bill_number, a_ar_category)

Canonical Join Paths

Customer → Account → Service → Meter → Usage → Bill:

ub_customers → utactcid → utsvcmst → utsvcmtr → utbilmtr → utbilsvc → utbilhdr

Utility Bill → AR Bill → Payment:

utbilhdr → ut_ar_reference → ar_all_bills → ar_receipt_all

Bill → Adjustment:

utbilhdr → utadjhdr → utadjsvc → utadjmtr